

MYC-C7Z010/20



Embedded SoC (ARM+FPGA) core board

Product Data Sheet

Overview
MYC-C7Z010/20 is an embedded system based on Xilinx Zynq-7010/7020 launched by Shenzhen Mir Technology Co., Ltd. It uses the latest Xilinx Zynq-7000 All Programmable SoC platform based on 28nm process flow, The processor and FPGA architecture are tightly integrated. The PS unit has the high performance and low power consumption of the dual-core ARM Cortex-A9 MPCore. The design can better meet various industrial needs. At the same time, Shenzhen Mir Technology Co., Ltd. provides a variety of mature hardware solutions, as well as a wealth of embedded operating systems and software Through the supporting design tools, it helps embedded software personnel to give full play to the advantages of software and hardware collaboration and achieve innovation beyond traditional architecture. New design.

Product Features
- Artix®-7 FPGA + dual ARM® Cortex-A8 cores, 866MHz main frequency
- 85,000 logic gates
- 1GB DDR3 SDRAM, 4GB eMMC, 16MB/32MB Qual SPI-Flash
- One Gigabit Ethernet PHY, one high-speed USB 2.0 PHY
- LCD controller, supports up to 1080p
- PCB technology: 10-layer board, immersion gold, lead-free, independent and complete ground plane
- Dimensions: 75mm x 55 mm
- Package size: 280 Pin 0.8mm pitch double row high density high speed PCB connector
- Supported operating system Linux

Application Areas
- Gaming peripherals, high-end toys
Home and industrial automation
Consumer medical devices
- printer
Smart charging system, smart vending machine
Weighing system, educational terminal

Project customization
- Select memory chips of different sizes according to customer needs
Tailor the system according to customer needs
- Assist in developing related drivers according to customer needs
Customized development of the baseboard according to the specific needs of customers



Version History

Version Number	illustrate	time
V1.0 Initial version		2015/05/11

Table of contents

Contents	1
Chapter 1 Overview	1
Chapter 2 Hardware Parameters	2
2.1 SoC Features.....	2
2.2 SoC BANK	3
2.2 Onboard Hardware Resources.....	4
Chapter 3 Pin Definition	5
3.1 Pin Location Diagram.....	5
3.2 PCB Connector.....	5
3.3 Pin Function Description Table.....	5
Chapter 4 Hardware Design Core Board Circuit	6
4.1 Power Supply.....	6
4.2 Boot Mode and JTAG Mode Selection.....	6
4.3 DDR.....	6
4.4 Storage.....	7
4.4.1 SPI Flash	7
4.4.2 eMMC.....	8
4.6 Ethernet.....	8
4.6 USB	8
Chapter 5 Mechanical Parameters	10
Chapter 6 Introduction to Supporting Development Board	11
Appendix 1 Contact Information	12
Appendix II After-sales Service and Technical Support	13

Chapter 1 Overview

MYC-C7Z010/20 is a Xilinx Zynq-7010/7020 based microcontroller launched by Shenzhen Mir Technology Co., Ltd.

It uses Xilinx's latest Zynq-7000 All-in-One based on 28nm process.

Programmable SoC platform, tightly integrating ARM processor and FPGA architecture, PS unit has dual-core ARM

The high performance and low power consumption of Cortex-A9 MPCore can better meet various industrial needs in design.

At the same time, Shenzhen Mir Technology Co., Ltd. provides a variety of mature hardware solutions and provides a wealth of embedded operating

The system and software resources, through the supporting design tools, help embedded software personnel to give full play to the advantages of software and hardware synergy.

Achieve innovative designs that go beyond traditional architectures.

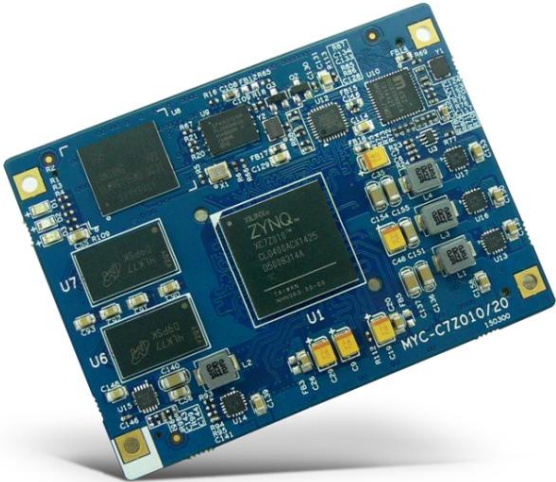


Figure 1-1 Product front view

There are two core board models to choose from, the difference is the SoC, as shown in the following table, XC7Z010 and XC7Z020

The differences are detailed in the next chapter.

Product Model	MYC-C7Z010	MYC-C7Z020
SoC	XC7Z010	XC7Z020
model	MYC-C7Z010-4E1D-667-C	MYC-C7Z020-4E1D-667-C
	MYC-C7Z010-4E1D-667-I	MYC-C7Z020-4E1D-667-I

Table 1-1

Chapter 2 Hardware Parameters

2.1 SoC Features

The core board uses Xilinx Zynq-7000 All Programmable SoC XC7Z010, which is the Zynq-7000 series

SoC, which tightly integrates ARM processor and FPGA architecture, has two ARM® Cortex™-A9 cores.

The FPGA consists of a processing core unit (PS) and a programmable logic unit (PL) composed of a Xilinx 7 series FPGA core Artix®-7.

In this series of chips, ARM has a more powerful computing speed than other processors on the market, and FPGA has a completely

Programming capability. The programmable capability of FPGA, combined with the I/O ports on the development board, can be suitable for a variety of application scenarios.

Note: The SoC model of the development board is XC7Z010 or XC7Z020. The two SoCs are pin-to-pin compatible and packaged

CLG400 (17 x 17 mm), XC7Z010 and XC7Z020 differ only in PL part, XC7Z020 has more logic cells

XC7Z010, suitable for more complex applications.

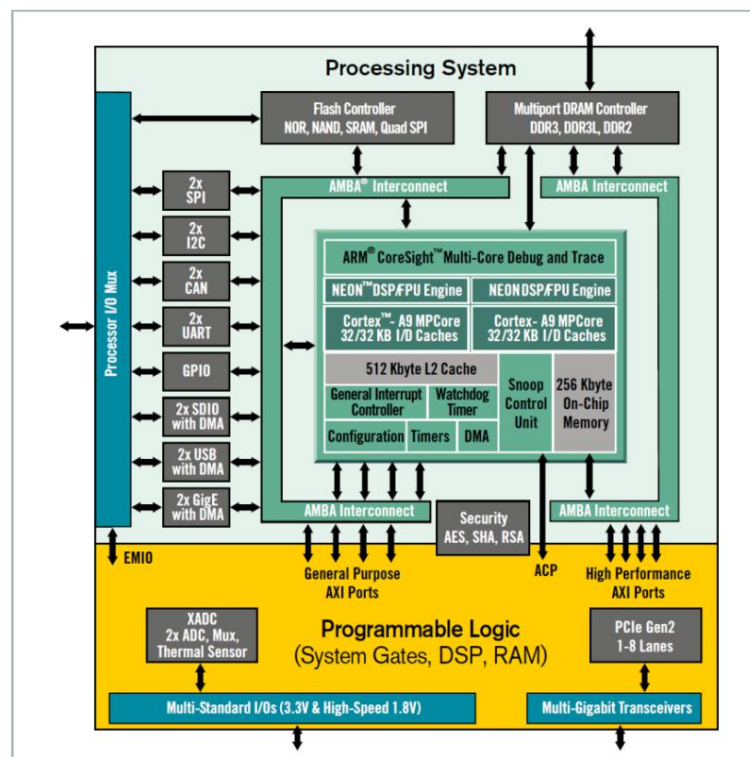


Figure 2-1 ZYNQ-7000 Series SoC Architecture

• Processor System Unit (PS)

• Processing core: Dual ARM® Cortex™-A9 MPCore™ with CoreSight™

• Maximum frequency: 866 MHz

• L1 Cache: 32 KB instruction cache, 32 KB data cache per core

- **L2 Cache** 512 KB
- On-chip memory: 256 KB
- External interface: DDR3, DDR3L, DDR2, LPDDR2
- Storage interface: 2x Quad-SPI, NAND, NOR
- **DMA** channels: 8 (4 of which are dedicated to PL)
- 2x UART, 2x CAN 2.0B, 2x I2C, 2x SPI, 4x 32b GPIO, 2x USB 2.0 (OTG),
2x Tri-mode Gigabit Ethernet, 2x SD/SDIO

• Programmable logic unit (PL)

	XC7Z010	XC7Z020
Logic Core	Xilinx 7 Series Artix®-7 FPGA	
Programmable Logic Unit	28,000 logic units, About 430,000 ASIC logic gates	85,000 logic units, Approximately 1.3 million ASIC logic gates
LUTs	17,600	53,200
trigger	35,200	106,400
RAM	240 KB	560 KB
DSP slice	80	220

Table 2-1

2.2 SoC BANK

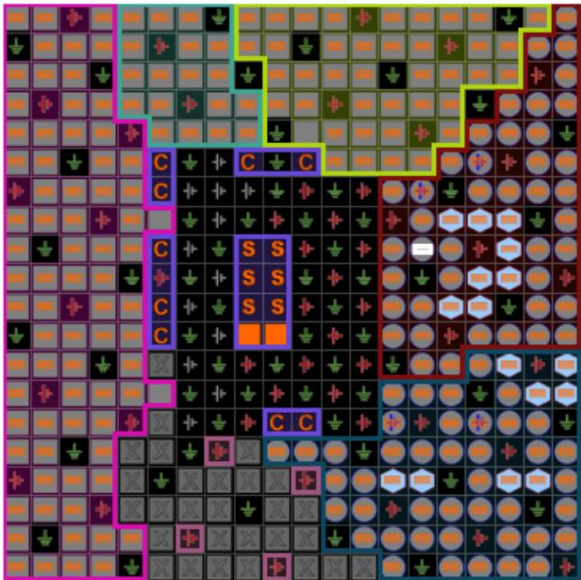


Figure 2-2 CLG400 BANKS

- **BANK 0:** JTAG, reset, analog signal
- **BANK 13:** Only available on XC7Z020
- **BANK 34:** PL unit, 50 pin (24 pairs of differential)
- **BANK 35:** PL unit, 50pin (24 pairs of differential)
- **BANK 500:** PS unit, MIO[0:15], 16pin, RGMII, USB, SDIO, UART
- **BANK 501:** PS unit, MIO[16:53] 38pin, QSPI, NAND Flash
- **BANK 502:** PS unit, DDR pin

2.2 Onboard Hardware Resources

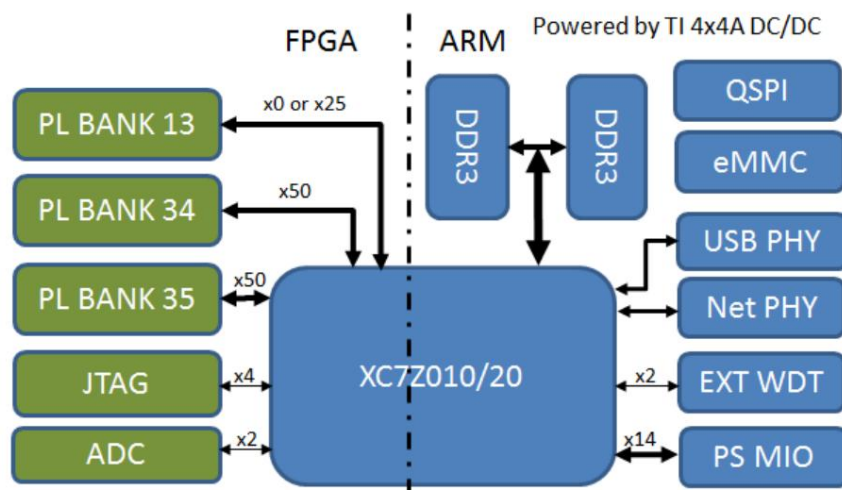


Figure 2-2 Core board onboard resource configuration

- 1GB DDR3 SDRAM
- 4GB eMMC
- Qual SPI-Flash 16MB/32MB
- 10/100/1000M Gigabit Ethernet interface
- External watchdog circuit
- A power indicator (blue), an FPGA programming completion indicator (red) and a user indicator (green color)

Chapter 3 Pin Definition

3.1 Pin location diagram

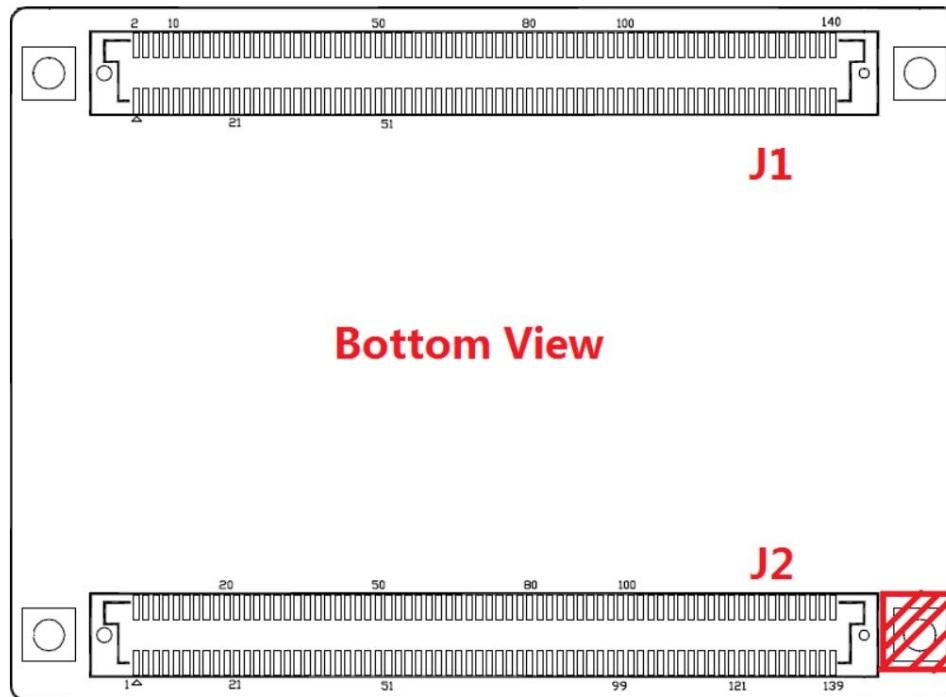


Figure 3-1 Core board pin diagram

3.2 PCB Connectors

The core board uses a 0.8mm double-row PCB connector, and the matching reference model specification is AMP's 5177983-6. The component package is provided in the information we provide, in the Allegro 16.6 format, which can be found in the Schematic folder.

3.3 Pin Function Description Table

For detailed definitions, please refer to the separate <<MYC-C7Z01020 Pin Description Table>> document.

Chapter 4 Hardware Design Core Board Circuit

4.1 Power Supply

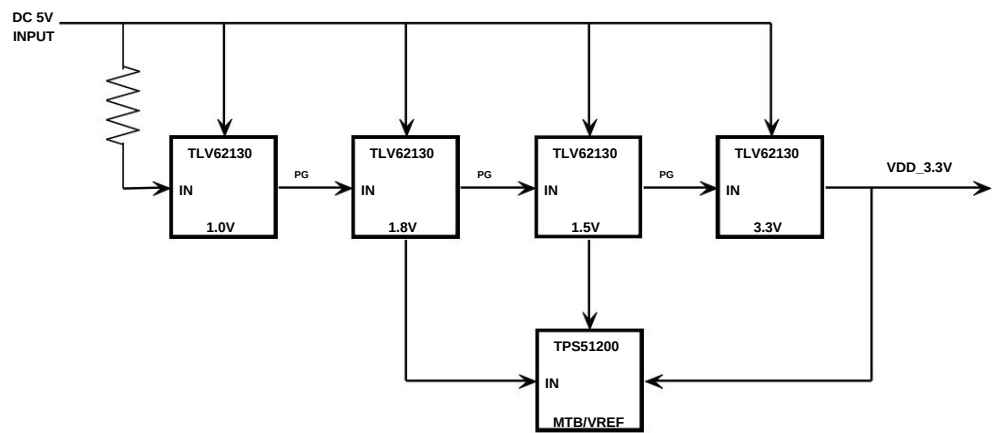


Figure 5-1

The above figure is the power block diagram of the development board. Through the Power good signal, it is cascaded in sequence. The power-on process is completed in the order of 1.0V->1.8V->1.5V->3.3V, and 3.3V rises last, providing a reset signal to the system. During the reset operation, the IO levels of the PL and PS pins are provided by the processor.

4.2 Boot mode and JTAG mode selection

The development board has two default boot modes to choose from: SD card boot and Quad-SPI boot.

The pull-up electronics of the pin are set.

Name	SD_BOOT_CONF_H S_JTAG_H	
AS Pin	J1 Pin-109	J1 Pin-111
SoC Pin	PS_MIO4	PS_MIO2
PS/PL JTAG Standalone Mode	X	H
QSPI Boot Mode	L	L
SD card boot mode	H	L

Table 5-1

4.3 DDR

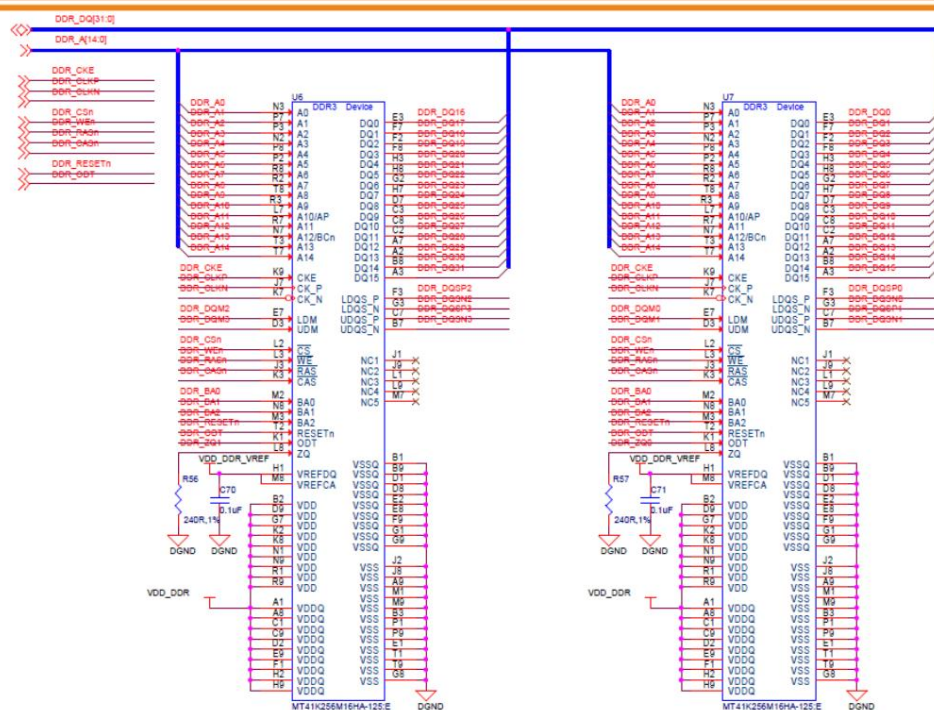


Figure 5-2

The development board includes two Micron MT41K256M16HA-125E DDR3 memory chips, forming a 256M

×32-bit interface, totaling 1 GB RAM. DDR3 memory connected to the SoC's PS DDR controller

The maximum supported speed on the physical port is 1066MT/s.

4.4 Storage

4.4.1 SPI Flash

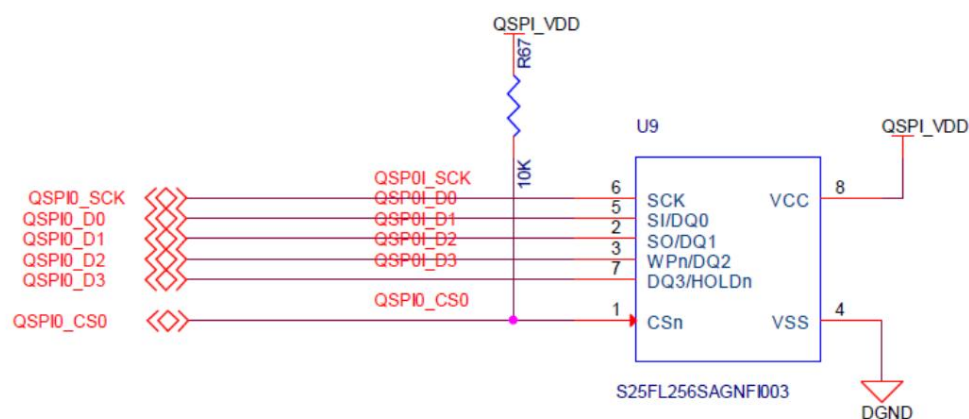


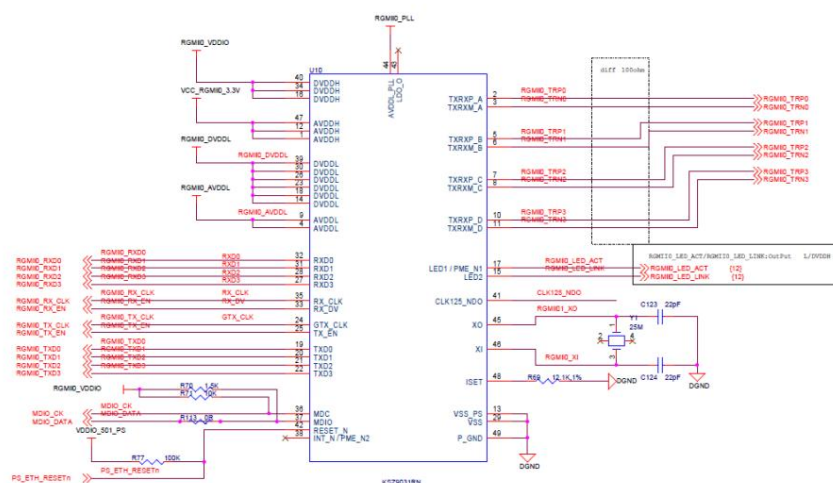
Figure 5-3

The development board comes with a Quad-SPI Flash that can be used to boot the SoC, initialize the PS and configure the PL unit.

4.4.2 eMMC

4GB eMMC large-capacity storage.

4.6 Ethernet

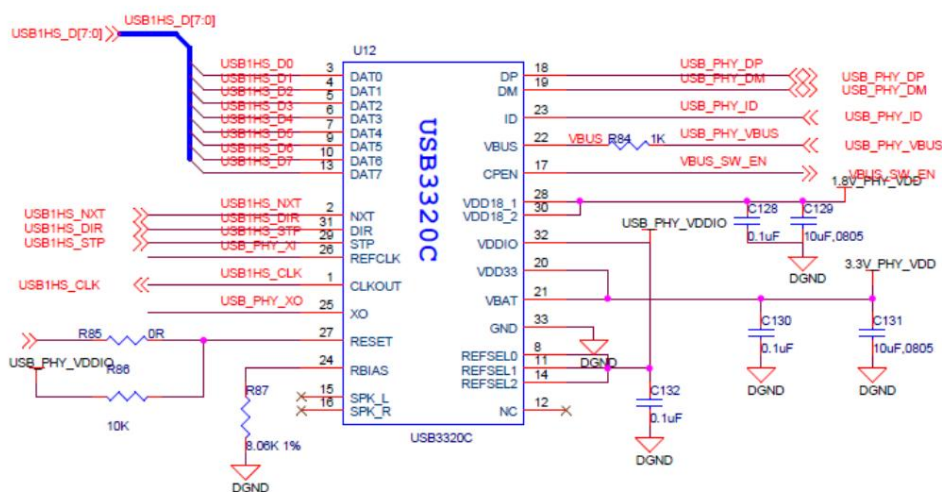


The PS unit of the Zynq-7000 SoC contains a Gigabit Ethernet MAC layer hardware controller, and the external

An Ethernet physical layer transmission chip. The development board uses KS29031RN as PHY and uses the RGMII interface to connect to a

Gigabit Ethernet port.

4.6 USB



The SoC is connected to a USB physical layer chip USB3320C from SMSC through the MIO of the PS unit.

It can be used as a USB 2.0 port. It can be used as both a USB host and a USB device.



It can connect USB devices such as USB flash drives and mice. When used as a USB device, it can connect to other hosts.

Acts as a USB network card or USB flash drive.

Chapter 5 Mechanical Parameters

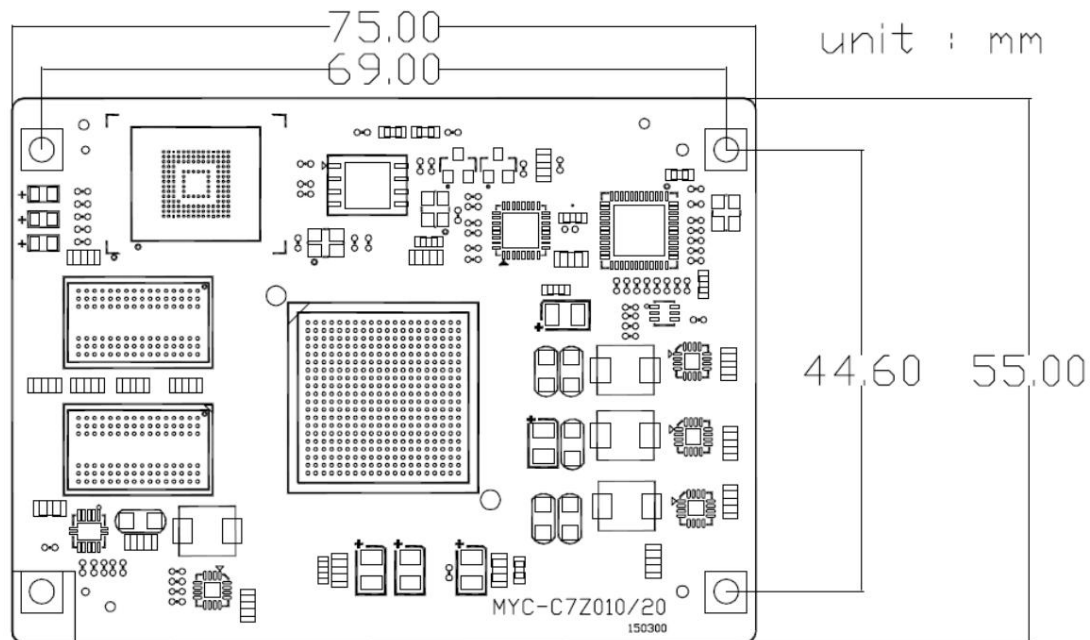


Figure 5-1 Core board mechanical dimensions

Working temperature: Industrial grade: -40~+85°C

Commercial grade: -20~+70°C

Ambient temperature: -50~+100°C

Ambient humidity: 20%~90%, non-condensing

Size: 75.0 mm x 55.0 mm, board thickness 1.0 mm, core board thickness about 4 mm

Mass: 12g

PCB specifications: 10-layer board design, immersion gold process, independent complete ground layer, lead-free process

Power input requirement: 5V 3.3V

System power consumption: 5W

Interface type: 280 Pin 0.8mm pitch double row high density high speed PCB connector

Chapter 6 Introduction to Supporting Development Board

MYD-C7Z010/20 is a development board for the MYC-C7Z010/20 series core board.

The motherboard is equipped with serial port, network port, MMC/SD/SDIO card interface, ADC interface, CAN and other interfaces, and supports Linux.

The information provided includes user manual, PDF baseboard schematic, external expansion interface driver, BSP source code package, development tools, etc. It provides developers with a complete software development environment, reduces product development cycle, and enables rapid product launch.



Figure 6-1 MYD-C7Z010/20

Appendix 1 Contact Information

Sales Contact

Website: www.myir-tech.com E-mail: sales.cn@myirtech.com

Shenzhen Headquarters

Responsible areas: Guangdong / Sichuan / Chongqing / Hunan / Guangxi / Yunnan / Guizhou / Hainan / Hong Kong / Macau
Tel: 0755-25622735 0755-22929657 Fax: 0755-25532724
Zip code: 518020
Address: Room 1306, Wensheng Center, Wenjin Plaza, No. 1010, Wenjin North Road, Luohu District, Shenzhen

Shanghai Office

Responsible area: Shanghai/ Hubei/ Jiangsu/ Zhejiang/ Anhui/ Fujian/ Jiangxi
Tel: 021-60317628 15901764611 Fax: 021-60317630
Zip code: 200062 Address: Room 1402, Building I, North Bank Changfeng, No. 106, Zhongjiang Road, Putuo District, Shanghai

Beijing Office

Responsible areas: Beijing/ Tianjin/ Shaanxi/ Liaoning/ Shandong/ Henan/ Hebei/ Heilongjiang/ Jilin/ Shanxi/ Gansu/ Inner Mongolia/ Ningxia
Tel: 010-84675491 13269791724 Fax: 010-84675491
Zip code: 102218 Address: Room 1009, Building 2, Runfeng Xinshang, Zhongtan Village, Dongxiaokou Town, Changping District, Beijing

Technical Support Contacts

Tel: 0755-25622735
Email: support@myirtech.com

Appendix II After-sales service and technical support

All Mir Technology products purchased directly from Mir Technology or from Mir Technology authorized agents

Products can enjoy the following benefits:

1. 6 months free warranty service period
2. Lifetime free technical support service
3. Lifetime maintenance service
4. Free software upgrade service for the purchased product
5. Free access to the software source code of the purchased product and some software source code developed by Mir Technology
6. You can directly purchase the main chip samples from Mir Technology, which is simple, convenient and fast; you don't need to go through the long process of purchasing from agents.

Long waiting period

7. From the date of purchase, you will become a permanent customer of Mir Technology and enjoy the right to purchase any software or hardware product of Mir Technology again

Preferential policies

8. OEM/ODM service

If any of the following situations occurs, you will not enjoy free warranty service:

1. Exceeding the free warranty service period
2. No product serial number or valid purchase receipt
3. Liquid ingress, moisture, mildew or corrosion
4. Failures and damages caused by impact, squeezing, falling, scratching, etc. that are not quality issues of the product itself
5. Failures and damages caused by unauthorized hardware modification, incorrect power-on, or incorrect operation
6. Failure and damage caused by irresistible natural factors

Product repair: If the product fails, is damaged or has other abnormalities during use, please

Call MirTech customer service department and communicate with engineers to confirm the problem to avoid unnecessary shipping costs caused by fault diagnosis errors.

Losses and cycle delays.



Repair cycle: After receiving the returned product, we will arrange engineers to test it on the same day, and we will repair it in the shortest time

Or replace and send back. The general fault repair cycle is 3 working days (from the date our company receives the item, excluding transportation time).

For products that cannot be repaired in a short period of time due to special faults, we will communicate with the user separately and confirm the repair schedule.

Expect.

Repair costs: During the free warranty period, no repair costs will be charged for failures caused by product quality problems;

For any fault or damage that falls within the scope of free warranty, we will communicate with the customer and confirm the repair cost after testing and confirming the problem.

We only charge for component materials, not repair service fees; for products that exceed the warranty period, we will charge a fee based on the actual degree of damage.

To determine the component material fees and maintenance service fees to be charged.

Shipping costs: When the product is under normal warranty, the shipping cost will be borne by the user. The cost of shipping back to the user after repair will be borne by our company.

The shipping costs of non-normal warranty products shall be borne by the user.